ENERGY EFFICIENCY MODULE-HEAT EXCHANGERS

HEAT EXCHANGER PERFORMANCE ASSESSMENT

Heat Exchangers can deteriorate over time due to off-design operation, fouling, scaling, or corrosion. This tool periodically assesses thermal duty, heat transfer coefficients (U), effectiveness (ϵ), and hydraulic pressure drops (ΔP) for both Shell & Tube and Plate Exchangers.

Metadata & Operational Configuration

Select Heat Exchanger Type:

Shell & Tube Heat Exchanger (STHE)

Design Data Inputs (from Test/Spec Reports)

Flow Configuration:

Counterflow

No. of Shell Side Passes (N)

1

No. of Tube Side Passes (M)

1

Total Design Heat Duty (kW)

25623.00

Overall Heat Transfer Coefficient U_d (kW/m²-K)

1.1780

Hot Fluid Side Pressure Drop Design (bar)

1.34

Cold Fluid Side Pressure Drop Design (bar)

0.95

Hot Fluid Temp Range (°C)

45.00

Cold Fluid Temp Range (°C)

25.00

Corrected LMTD Design (°C)

83.80

Operation Data Inputs

Hot Fluid Flow Inlet (kg/hr)
719800.00

Hot Fluid Temp Inlet (°C)
145.00

Hot Fluid Pressure Inlet (bar g)
4.10

Cold Fluid Flow Inlet (kg/hr)
881150.00

Hot Fluid Temp Outlet (°C)
102.00

Hot Fluid Pressure Outlet (bar g)
2.80

Cold Fluid Temp Inlet (°C)
25.50

Cold Fluid Pressure Inlet (bar g)
6.20

Cold Fluid Temp Outlet (°C)
49.00

Cold Fluid Pressure Outlet (bar g)
5.10

Fluid Properties

Hot Fluid Specific Heat Capacity (kJ/kg-°C)
2.84700

Latent Heat of Condensation (kJ/kg)
0.00000

Cold Fluid Specific Heat Capacity (kJ/kg-°C)
4.18700

Latent Heat of Vaporisation (kJ/kg)
0.00000

Heat Transfer Area (A) Calculation

How do you want to provide the Heat Transfer Area (A) m^2?

- (•) Calculate A (Tube Geometry)
- ( ) Input A Manually

Outer Diameter of Tubes (mm)
25.40

No. of Tubes
460

Length of Tubes (mm)
7211

Final Calculated Heat Transfer Area (A): 264.6896 m^2

How do you want to provide the Correction Factor (F)?

- (•) Calculate F
- ( ) Input F Manually

Assessment Results

	Parameters	Units	Operation Data	Design Data
0	Heat Exchanger Type	-	Shell & Tube Heat Exchanger (STHE)	Shell & Tube Heat Exchanger (STHE)
1	Duty (Q)	kW	24477.40	25623.00
2	Hot Fluid Pressure Drop ΔPh	bar	1.30	1.34
3	Cold Fluid Pressure Drop ΔPc	bar	1.10	0.95
4	Temp Range Hot Fluid	°C	43.00	45.00
5	Temp Range Cold Fluid	°C	23.50	25.00
6	Corrected LMTD	°C	83.88	83.80
7	Heat Transfer Coefficient U	kW/m²-K	1.103	1.178

Effectiveness (ε)

0.3598

Capacity Ratio (r)

0.5555

Fouling Resistance (Rd)

0.0581 m^2K/kW

AI Performance Diagnostic Summary

Generate AI Assessment Summary

Diagnostic Summary (Shell-&-Tube Heat Exchanger)

- **Thermal performance:** The operating overall heat-transfer coefficient ($U = 1.103\ kW\,m^{-2}\,K^{-1}$) is **≈6 % lower** than the design value ($U_{design} = 1.178\ kW\,m^{-2}\,K^{-1}$). The corrected LMTD is essentially unchanged (83.88 °C vs. 83.80 °C), indicating that the reduced U is the primary driver of the 5 % shortfall in duty (24 477 kW vs. 25 623 kW). The calculated fouling resistance ($R_f \approx 0.058\ m^2\,K\,kW^{-1}$) is consistent with moderate tube-side fouling.

- **Hydraulic performance – hot side:** Pressure drop has risen from the design 1.34 bar to 1.30 bar **(≈3 % lower)**, which is acceptable and suggests no significant blockage on the hot-side flow path.
- **Hydraulic performance – cold side:** Pressure drop has increased from the design 0.95 bar to 1.10 bar **(+16 %)**, indicating a developing restriction on the cold-side side (likely tube-side fouling, scaling, or partial blockage).
- **Effectiveness:** The thermal effectiveness (ε = 0.3598) is low for a counter-flow configuration (F = 0.977), confirming that the heat-transfer area is not being fully utilized because of the elevated fouling resistance.

Comparison of Thermal vs. Hydraulic Degradation

Parameter	Design	Operating	% Change	Interpretation
Overall U-value ($kW\,m^{-2}\,K^{-1}$)	1.178	1.103	**-6 %**	Fouling or deposit layer on tube walls reducing conductive/convective heat transfer.
Hot-side ΔP (bar)	1.34	1.30	**-3 %**	Within tolerance; no major flow restriction.
Cold-side ΔP (bar)	0.95	1.10	**+16 %**	Indicative of tube-side fouling, scaling, or partial blockage; the dominant hydraulic penalty.

The thermal penalty (-6 %) is modest compared with the hydraulic penalty on the cold side (+16 %). The latter will increase pumping power and may exacerbate fouling by reducing velocity, further degrading U.

Recommended Maintenance Actions (STHE-specific)

1. **Mechanical cleaning – tube rodding & bundle pull-out**
   - Perform a full-bundle removal and rod the tubes with appropriate diameter rods (e.g., 0.8 × tube OD) to dislodge scale and particulate deposits.
   - Inspect tube integrity (wall thickness, ovality) after removal; replace any tubes exceeding allowable wear criteria.
2. **Chemical cleaning (CIP) of the cold-side circuit**
   - Circulate a high-pH (NaOH) or acidic (HCl) cleaning solution at the recommended temperature (≈80 °C) for 2–4 h, followed by thorough flushing with de-ionized water.
   - Select cleaning chemistry based on the fouling type (organic vs. inorganic scaling) identified during visual inspection.
3. **Hydro-blasting / high-velocity water jetting (optional)**
   - If chemical cleaning alone does not restore ΔP to design levels, apply water-jetting (≥150 bar) through the tube bundle to remove stubborn deposits without damaging tube material.
4. **Post-cleaning verification**

- Re-measure hot- and cold-side pressure drops and conduct a performance test (U-value, LMTD) to confirm that the thermal duty returns to ≥ 95 % of design.
- Update the fouling factor (R_f) in the heat-exchanger rating to refine future maintenance intervals.

Implementing the above actions on a scheduled basis (e.g., every 12–18 months, or when ΔP exceeds +10 % of design) will mitigate fouling growth, restore hydraulic balance,